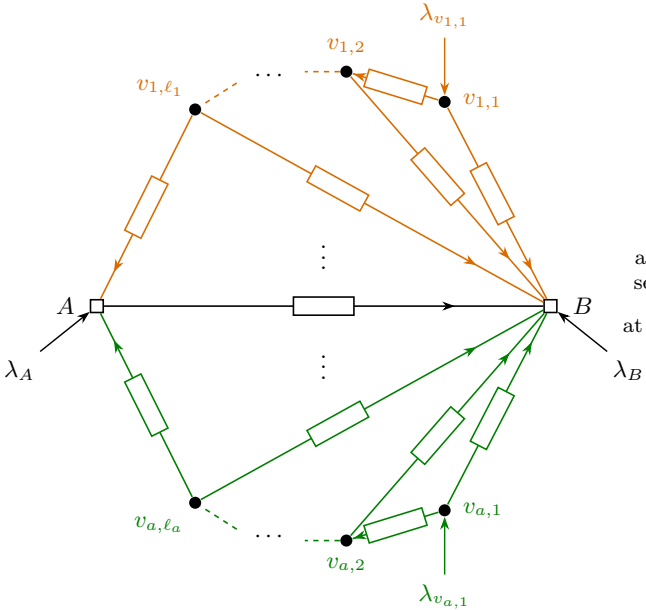
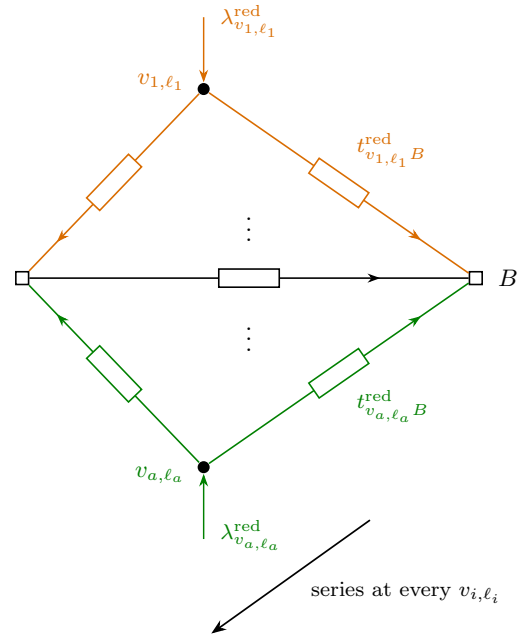
(a) Spider: arms meet at A, every arm vertex wired to B

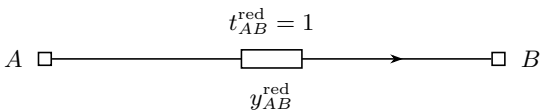


along each arm: series + parallel at $v_{i,1}, \ldots, v_{i,\ell_i-1}$

(b) Each arm reduced to its last vertex



(d) The unit resistor



$$y^{\text{red}}_{AB} = \lambda_A + \textstyle\sum_i \big(\prod_{k=1}^{\ell_i} r_{i,2k-1}\big)\lambda_{v_{i,1}}$$

one parallel merge

(c) One resistor per arm, all between A and B

